

Optimizing bin size in histogramming data

There is no optimal bin size for all distributions and all histograms. The choice may depend on what features of the data you choose to emphasize. Nevertheless, starting with Sturges [1] there have been attempts to establish some rough guidelines of which data scientists should be aware.

Suppose I have a data set consisting of N measurements of an IID random variable x , with values $x_{min} < x < x_{max}$. I wish to sort the x_i into uniform bins of width Δx .

Sturges formula, derived by assuming an approximately normal distribution of x , recommends a bin width of approximately

$$\Delta x = \frac{(x_{max} - x_{min})}{1 + \log_2 N}$$

A more recent formulation, also optimized for normally distributed data, is due to Scott [2]:

$$\Delta x = \frac{3.49\sigma_x}{N^{1/3}}$$

Doane [3] extended Sturges formula to the case of non-normally distributed data as follows:

$$\Delta x = \frac{(x_{max} - x_{min})}{1 + \log_2 N + \log_2 (1 + |g_1|/\sigma_{g_1})}$$

where g_1 is the estimated skewness and σ_{g_1} is given by

$$\sigma_{g_1} = \sqrt{\frac{6(N-2)}{(N+1)(N+3)}}$$

Application of the above to Homework 2, Problem 3

In Homework 2, Problem 3 we are asked to sort $N = 4,201$ exponentially distributed data points between $0 < t < 8 \mu\text{sec}$ into a histogram for the purpose of performing a least-squares fit.

$$f(t|\tau) = \frac{1}{\tau} e^{-t/\tau}$$

The expected value of the lifetime τ is approximately $2 \mu\text{sec}$. Below, I summarize the binning recommendations for the three formulae along with the “professor’s eyeball” recommendation.

method	bin size
Sturges	$0.614 \mu\text{sec}$
Scott	$0.433 \mu\text{sec}$
Doane	$0.425 \mu\text{sec}$
JB’s eyeball	$0.320 \mu\text{sec}$

References

- [1] Sturges, H. A., “The choice of a class interval”, J. Am. Stat. Assn., **21** (153) pp 65–66 (1926).
- [2] Scott, D. W., “On optimal and data-based histograms”, Biometrika **66** (3) pp 605–610 (1979).
- [3] Doane, D. P., “Aesthetic frequency classification”, Am. Stat. **30** 181–183 (1976).